

InuaBiz System Architecture & Commercial Strategy

Technical Specification & Enterprise Deployment Blueprint

1. Architectural Foundation & Tenant Isolation

Core Platform Model

Built as a unified, high-performance multi-tenant SaaS that caters to single-branch vendors, multi-shop organizations, and standalone white-label enterprise clients.

Database Routing & Isolation

Shared Multi-Tenant Instance: Single vendors and multi-branch organizations operate on a shared Supabase PostgreSQL backend using Row-Level Security (RLS) driven by *organization_id* and *shop_id*.

Dedicated Enterprise Instance: Enterprise clients requesting custom isolation receive a dedicated Supabase/PostgreSQL environment. The application layer routes API calls to their isolated database URI while sharing the core codebase logic.

Niche-Specific UI & Module Mapping

Dynamic module loading based on the vendor's selected business vertical (e.g., retail, electronics, pharmacy, agriculture). Master vendors can assign distinct niches to different shops under the same organization, dynamically altering UI components, checkout schemas, and inventory fields per location.

2. Master Vendor & Multi-Shop Management

Centralized Control & Hierarchy

Master Vendor Level: Full administrative control over organization-wide settings, user roles, centralized reporting, and subscription management.

Shop/Master Level: Operational access restricted to specific locations, inventory nodes, and local point-of-sale terminal configurations.

Unified Onboarding Flow

- 1. Organization Creation:** Master vendor registers the primary entity and billing details.
- 2. Shop Provisioning:** Master vendor adds new store locations and selects individual business domain niches for each.
- 3. Role Assignment & Access Control:** Granular RBAC configuration to assign managers, cashiers, and inventory clerks per location.

3. Commercial Framework & Hybrid Pricing Strategy

Standard Multi-Tenant Pricing Structure

Base Subscription Fee: Consolidated billing handled exclusively at the Master Vendor level.

Per-Shop Add-On Fee: Scaled pricing based on the total number of active shop nodes under a single master account.

Automated Feature Gating: InuaBiz M-Pesa automated billing integration gates access organization-wide if subscription renewals are missed, removing individual per-shop collections.

Standalone Enterprise Licensing Model

Upfront Licensing & Customization Fee: One-off fee for custom software builds, white-label UI branding, and dedicated infrastructure deployment.

Recurring Maintenance & Operations Fee: Monthly or annual contract covering dedicated cloud hosting, database administration, continuous platform updates, and SLA-backed support.

Commercial Models Comparison

Model	Target Client	Billing Structure
Multi-Tenant SaaS	Standard Vendors / Multi-Shop Organizations	Base Plan + Per-Shop Add-On Fee (M-Pesa Consolidated Billing)
Enterprise Custom Build	Standalone System Clients	Upfront License Fee + Monthly SLA Maintenance/Hosting Contract

4. Enterprise Operations & Admin Oversight

Super Admin Operations Dashboard

Global Tenant Monitoring: Real-time operational tracking of system health, active database connections, and usage metrics across shared and dedicated enterprise instances.

Automated Feature Rollouts: CI/CD pipeline deployment pushing feature updates simultaneously to the multi-tenant core and white-labeled enterprise codebases.

Licensing & Contract Management: Systemic tracking of enterprise SLAs, recurring maintenance agreements, and custom feature flags.